Lecture 09 - Exercises

These exercises accompany Lecture 09, "Correlation and Regression."

Part A is by hand. It builds the mechanics: variables, slopes, fitted values, residuals, least squares, correlation, and extrapolation.

Part B uses Python with the official exercise workbook. The goal is to estimate regressions, compute residuals, compare fits, and interpret output carefully.

By-hand solutions are provided separately. Python solutions are in the solutions notebook.

Part A - By-Hand Practice

Exercise 1 - Identify the Regression Question

For each situation, identify the dependent variable, the independent variable, and the main purpose: explanation or prediction.

Situation	Question
A	Does advertising spending help explain sales?
B	Does price help predict demand?
C	Does delivery time help predict customer satisfaction?
D	Does age help predict annual spending?

Exercise 2 - Slope and Intercept

A fitted regression line is

```
y_hat=12+3x.
```

1. What is the intercept?
2. What is the slope?
3. What is the fitted value when x=4?
4. Interpret the slope in words.
5. When might the intercept be meaningful? When might it be only a mathematical anchor?

Exercise 3 - Fitted Values and Residuals

A model predicts revenue from investment:

```
Revenue_hat=25+2 x Investment.
```

Compute the fitted value and residual for each observation. Use residual = observed - fitted.

Observation	Investment	Revenue
1	10	46
2	15	50

3	20	70
4	25	76

Exercise 4 - Compare Two Candidate Lines

Use the same data as Exercise 3.

```
y_hat_A=20+2x,

y_hat_B=30+1.5x.
```

For each line:

1. compute residuals;
2. square the residuals;
3. compute the total squared error;
4. choose the better line under least squares.

Exercise 5 - Correlation Interpretation

Interpret each correlation.

Case	Correlation
A	0.92
B	0.15
C	-0.70
D	0.00

Then explain why correlation does not prove causation.

Exercise 6 - From Correlation to Slope

Suppose r=0.80, s_X=5, and s_Y=20.

1. Compute

```
b_1=r(s_Y / s_X).
```

2. Interpret the slope.
3. Explain how the slope differs from correlation.

Exercise 7 - Interpolation or Extrapolation?

A regression model is estimated from investment values between 10 and 25.

Classify each prediction.

1. Investment = 12.
2. Investment = 20.
3. Investment = 30.
4. Investment = 80.

Why is extrapolation riskier?

Exercise 8 - Residual Diagnostics in Words

Explain what each pattern may suggest.

1. Residuals are centered around zero with no visible pattern.
2. Residuals form a U-shape against fitted values.
3. Residuals become more spread out when fitted values are high.
4. One observation has a much larger residual than the rest.

Exercise 9 - Underfitting and Overfitting

For each case, say whether the model is likely underfitting, overfitting, or reasonably balanced.

1. A straight line is used for a clearly curved relationship.
2. A very flexible curve passes almost exactly through every point in a noisy sample.
3. A simple model captures the main pattern and has similar error on training and new data.
4. A model performs extremely well on training data and badly on new observations.

Part B - Python Applications

Use:

`data_lecture9_exercises.xlsx`

The Excel workbook contains the regression practice datasets for these exercises. Use sheet `data1` for Exercises 10-14, and sheets `data2`, `data3`, and `data4` for Exercise 15.

Exercise 10 - Load the Regression Workbook

1. Load `data_lecture9_exercises.xlsx`.
2. Read the sheet `data1`.
3. Inspect the first rows and column names.
4. Treat `Revenue` as the dependent variable and `Cost` as the independent variable.
5. Report the sample size, mean cost, and mean revenue.

Exercise 11 - Estimate a Simple Regression by Formula

Using `data1`, estimate

```
Revenue_i=b_0+b_1Cost_i+e_i.
```

Compute:

1. the slope using covariance divided by variance;
2. the intercept;
3. the correlation between cost and revenue;
4. fitted values;

5. residuals.

Exercise 12 - Goodness of Fit

Using the regression from Exercise 11:

1. estimate the same regression using `statsmodels`;
2. read the coefficient of determination R^2 from the regression output;
3. compare R^2 with r^2, the squared correlation between cost and revenue;
4. interpret R^2 without making a causal claim.

Exercise 13 - Slope Uncertainty

Using the `statsmodels` output from Exercise 12:

1. read the standard error of the slope;
2. read the t-statistic for testing whether the slope is zero;
3. read the 95% confidence interval for the slope;
4. explain what the confidence interval means in context;
5. write a careful interpretation.

Exercise 14 - Interpolation and Extrapolation

Using the fitted line from `data1`:

1. find the minimum and maximum observed cost;
2. predict revenue for cost values inside the observed range;
3. predict revenue for a cost value far outside the observed range;
4. label each prediction as interpolation or extrapolation;
5. explain the risk in the extrapolated prediction.

Exercise 15 - Nonlinear Data and Linear Fit

Read the sheets `data2`, `data3`, and `data4`.

For each sheet:

1. identify the dependent and independent variables;
2. estimate a simple linear regression;
3. report the slope, intercept, and coefficient of determination R^2;
4. explain why R^2 alone does not prove the model is appropriate;
5. identify which dataset looks most problematic for a straight-line model.